

Zhao's method, and why it does not reach a single prime

What the literature proves about $v_2(L(1)/\Omega)$ for $y^2 = x^3 + Dx$, and why the Fermat primes sit outside it

third note on III at the Fermat primes

1. The state of the art, and it is our exact family

The previous note reduced everything to $v_2(L(E_p, 1)/\Omega) = 2k - 3$ and suggested looking at the literature on 2-adic valuations of central L -values in CM families. That literature exists, it uses **Zhao's method**, and one paper treats precisely the curves in question.

1.1. Zhao's method in one paragraph

For a CM elliptic curve the central value $L(1)/\Omega$ can be written as an **explicit finite sum** of special values of the Weierstrass function $\mathcal{P}$ — an expression going back to Birch and Swinnerton-Dyer. Zhao's idea is to estimate the 2-adic valuation of that sum by **induction on the number of distinct prime divisors of the twisting parameter D** : one relates the sum for D to the sums for the divisors D_T of D , and each prime factor contributes a fixed amount to the valuation. The output is always a lower bound of the shape

$$v_2(L(1)/\Omega) \geq \alpha \cdot r(D) + \beta,$$

with $r(D)$ the number of prime divisors.

1.2. Nomoto's theorem

Theorem (Nomoto, [arXiv:2207.10380](https://arxiv.org/abs/2207.10380), Thm 1.1). Let $K = \mathbb{Q}(i)$ and let $E_{-D} : y^2 = x^3 + Dx$ over K , with $D \in \mathcal{O}_K$ quartic-free and congruent to 1 modulo $2 + 2i$. Let ψ_{-D} be the associated Hecke character and let $L_2(\bar{\psi}_{-D}, s)$ be its L -function with the Euler factor at $(1 + i)$ removed. If $D \notin K^{\times 2}$ then

$$v_2\left(\frac{L_2(\bar{\psi}_{-D}, 1)}{\Omega}\right) \geq \frac{r(D) - 2}{2},$$

where $r(D)$ is the number of distinct primes dividing D and $\Omega = 2.6220575\dots$ is the least positive real period of $y^2 = x^3 - x$. For $D \in K^{\times 2}$, Zhao's bound is $(2r(D) - 3)/2$.

This is **exactly** our family — $y^2 = x^3 + Dx$, quartic-free D , CM by $\mathbb{Z}[i]$, the same period. So the question is whether it says anything about $D = p$.

2. Why it says nothing here

It does not, and the reason is structural rather than technical.

$r(D)$ counts primes of $\mathcal{O}_K = \mathbb{Z}[i]$, not of $\mathbb{Z}$. Our $D = p$ is a rational prime $\equiv 1 \pmod{4}$, so it **splits**: $p = \pi\bar{\pi}$ and $r(D) = 2$. The bound reads

$$v_2 \geq \frac{2 - 2}{2} = 0,$$

which is vacuous; Zhao's square case would give $1/2$. Meanwhile the truth, for $p = 2^{2^k} + 1$, is $v_2 = 2k - 3$, which grows without bound.

So the Fermat primes are **extremal in the direction the theory does not control**. Every bound in this line grows with the number of prime factors of D ; our family has the minimum possible number

of them — one rational prime, two Gaussian primes, fixed forever — and the valuation grows anyway. Nothing in the Zhao framework is shaped to see that.

3. Two candidate mechanisms, both eliminated

The same paper supplies the one local invariant at 2 that these arguments turn on, and it is constant on our family.

Proposition (Nomoto, Prop. 2.1). E_{-D} has good reduction at $(1+i)$ if and only if $(i/D)_4 = i$, and the valuations with and without the Euler factor at $(1+i)$ differ by $1/2$ exactly in that case.

For $D = p$ a rational prime the symbol is computable in closed form:

$$(i/p)_4 = (i/\pi)_4(i/\bar{\pi})_4 = i^{(p-1)/4} \cdot i^{(p-1)/4} = i^{(p-1)/2},$$

which for $p \equiv 1 \pmod{8}$ is 1. So $(i/p)_4 = 1 \neq i$ for **every** prime in our family: E_p always has bad reduction at $(1+i)$, there is never a half-shift, and the invariant is constant where the answer is not.

A second guess, that the answer is driven by how 2-divisible $p-1$ is — natural, since $p-1 = 2^{2^k}$ is as 2-divisible as it gets — also fails:

p	$v_2(p-1)$	b	m	$v_2(L(1)/\Omega)$
193	6	12	1	1
577	6	24	2	3

Same $v_2(p-1)$, different answer. Together with the counterexample 4177 versus 4937 of the previous note — same b , both of the form $x^2 + 256y^2$, different m — three of the four natural invariants at 2 are eliminated: the quartic symbol $(i/p)_4$, the octic residue character of 2, the 2-divisibility of $p-1$, and the shape of b .

4. What does survive

One statement, from the previous note, and it now sits naturally in this framework, since quartic residue symbols are exactly the currency of these papers:

For every rank-0 prime $p \equiv 1 \pmod{8}$ tested (82 of them),

$$m \text{ even} \iff 2 \text{ is a quartic residue mod } p \iff 8 \mid b,$$

where $L(1)/\Omega = 2m^2$ and $p = a^2 + b^2$ with a odd. For a Fermat prime $b = 2^{2^{k-1}}$, so this holds exactly when $k \geq 3$ — which is exactly when m stops being odd.

That fixes $v_2(L(1)/\Omega) = 1$ versus ≥ 3 . It is the first rung, and the ladder above it is what remains unknown.

5. Conclusion

What we can now say precisely. The pattern $\text{III}(E_p) \cong (\mathbb{Z}/2^{k-1})^2$ has three layers. The **shape** is a theorem for any $p \equiv 1 \pmod{8}$ with $L(1) \neq 0$, needing no BSD (first note). The **parity** of the square root m is Gauss's quartic residue criterion (second note). The **height** $m = 2^{k-2}$ is open, and — this is the point of this note — it is open in a regime the existing machinery does not address: all known lower bounds grow with the number of prime divisors of D , and here that number is pinned at its minimum while the valuation grows.

What would settle it. Either an exact evaluation, rather than a lower bound, of the Birch–Swinnerton-Dyer finite sum for $D = p$ prime — the sum is explicit, so this is a concrete if delicate 2-adic computation — or an identification of m as a coefficient of a half-integral weight form for the quartic-twist family, which is what the doubly logarithmic growth $\log_2 m = \log_2 \log_2 b - 1$ is pointing at.

Sources. K. Nomoto, *Lower bound for the 2-adic valuations of central L -values of elliptic curves with complex multiplication*, [arXiv:2207.10380](#) — treats $y^2 = x^3 + Dx$ over $\mathbb{Q}(i)$, quartic-free D ; Theorem 1.1 and Proposition 2.1 quoted above.
T. Adachi, K. Nomoto, R. Shii, *The 2-adic valuations of the algebraic central L -values for quadratic twists of weight 2 newforms*, [arXiv:2403.11474](#) — the quadratic-twist analogue, with bounds $\mathfrak{v}_m \cdot r + c$ linear in the number of prime factors.
C. Zhao, *A criterion for elliptic curves with lowest 2-power in $L(1)$ (I), (II)*, Math. Proc. Cambridge Philos. Soc. — the original method.